

Minh-Thanh Nguyen

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EDUCATION

Trinity College Dublin, The University of Dublin

Ph.D. in Computer Science

Dublin, Ireland

03.2026 – 02.2030

Hanoi University of Science and Technology (HUST)

B.Eng. in Electronics and Telecommunications. GPA: 3.32/4.0 (very good)

Hanoi, Vietnam

09.2021 – 08.2025

PROJECTS

Energy aware resource allocation in Open RAN using multi-agent DRL

03.2026 – 02.2030

PhD Student

ICCLab, TCD

- Formulate optimization model for allocating resources in Open RAN with energy considerations.
- Build a simulation framework for comparing performance of different algorithms for various Open RAN resource allocation with green energy consumption problems.
- Build a multi-agent DRL system which solve the problem of energy aware resource allocation in Open RAN.

Deploy 5G network slicing testbed

01.2025 – 02.2026

Research assistant

ANSALab, HUST

- Deploy and administration a private cloud computing system.
- Deploy a 5G networks with basic components with end-to-end slicing capability.

Efficient resource allocation for 5G network slicing

09.2023 – 10.2025

Research assistant

ANSALab, HUST

- Build a simulation framework for comparing performance of different algorithms for various variant of network slice embedding problems.
- Formulate optimization model for different variants of network slice embedding problem.

SKILLS

Programming: Python, C/C++, C#, JavaScripts, Lua, Bash, PowerShell

Technical: OpenStack, Kubernetes, Basic Linux Server Administration

Languages: Vietnamese (native), English (IELTS 7.0)

Writings: Microsoft Word, Microsoft PowerPoint, L^AT_EX

AWARDS

Best Graduation Thesis Award *Hanoi University of Science and Technology*

Jul. 2025

First Prize at 2025 Student Research Competition *Hanoi University of Science and Technology*

Jun. 2025

Best Poster Award *Asian Internet Engineering Conference 2023, Hanoi*

Dec. 2023

PUBLICATIONS

Journals

[J1] Q.-T. Luu, **M.-T. Nguyen**, T.-A. Do, M. Kieffer, T. H. Nguyen, H. T. Nguyen, and V.-D. Nguyen, “Network slicing with flexible vnf order: A branch-and-bound approach,” *IEEE Transactions on Network and Service Management*, vol. 23, pp. 3805–3818, 2026.

[J2] Q.-T. Luu, D.-M. Tran, **M.-T. Nguyen**, M. Kieffer, D. T. Hoang, T.-H. Nguyen, H.-T. Nguyen, and V.-D. Nguyen, “Network slice embedding with flexible configurations in 5g networks and beyond,” *IEEE Networking Letters*, pp. 1–1, 2026.

Conferences

[C1] Q.-T. Luu, **M.-T. Nguyen**, T.-H. Nguyen, M. Kieffer, V.-D. Nguyen, Q.-L. Luu, and T.-T. Nguyen, “Admission Control and Embedding of Network Slices with Flexible VNF Order,” in *2024 20th International Conference on Network and Service Management (CNSM)*, 2024, pp. 1–5.

[C2] **M.-T. Nguyen**, Q.-T. Luu, T.-H. Nguyen, D.-M. Tran, T.-A. Do, K.-H. Do, and V.-H. Nguyen, “Accelerating Network Slice Embedding with Reinforcement Learning,” in *2024 Tenth International Conference on Communications and Electronics (ICCE)*, 2024, pp. 78–83.

REFERENCES

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